BIO 005 Human Physiology · Week 10 notes · Dr. Sharilyn Rennie

Heart Physiology

The anatomy you need to follow the physiology, smooth and cardiac muscle, the adrenal medulla, the heart's electrical system and the EKG, one cardiac cycle from start to finish, cardiac output and how the body controls it, and what happens when the pump fails. Week 10 competencies 1 to 21.

Prefer paper? Print or save these notes as a two-column PDF.

How to use these notes. Watch the lecture first, then read these notes. The section letters match the lecture, and each part links Dr. Rennie's videos for it.

Your textbook: Silverthorn Chapter 15, Cardiovascular Physiology. **Using OpenStax instead?** Anatomy and Physiology 2e, Chapter 19, The Cardiovascular System: The Heart.

Need-to-know anatomy

In your textbook: Silverthorn Chapter 15, the part on the structure of the heart.

You only need a few structures to follow the physiology in these notes: where the heart sits, its four chambers and why their walls differ, the four valves and what makes them open and close, and the path blood takes through the heart.

Where the heart sits

- The heart is about the size of a closed fist. It rests on the diaphragm in the mediastinum, near the middle of the chest, with about two thirds of its mass to the left of the midline.
- The apex is the pointed tip, formed by the left ventricle. It points down, forward and to the left.
- The base is the broad upper back part, formed mostly by the left atrium. The great vessels attach here.

The pericardium and the heart wall

Pericardium

The sac around the heart. The tough outer fibrous layer anchors the heart and keeps it from overstretching. The thin inner serous layer is folded into two layers with a few mL of pericardial fluid between them, so the heart can beat with very little friction.

Epicardium

The outer layer of the heart wall. It is the visceral layer of the serous pericardium, and the coronary vessels run in it.

Myocardium

The middle layer, made of cardiac muscle. It is about 95% of the wall and does the pumping.

Endocardium

A smooth inner lining of endothelium that covers the chambers and valves. It is continuous with the lining of the blood vessels and keeps friction low as blood moves through.

Front view of the heart cut open to show the four chambers, the four valves, the chordae tendineae and papillary muscles, the interventricular septum, and the three layers of the heart wall.

The inside of the heart from the front: the chambers, the valves, and the three layers of the wall.
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Why the chamber walls are different thicknesses

- The atria only push blood a short distance into the ventricles, against low pressure, so their walls are thin, about 2 to 3 mm.
- The right ventricle pumps a short distance to the lungs, against low pressure, so its wall is about 4 to 5 mm.
- The left ventricle pumps blood to the whole body, against the highest pressure, so its wall is the thickest, about 10 to 15 mm.
- Even though the two ventricles push against very different pressures, they eject the same volume of blood with each beat. The output of one side becomes the input of the other.

The valves open and close because of pressure

Watch: ▶ Lub dub: how the valves open and close (Khan Academy)

Valves keep blood moving one way. No nerve or muscle opens or closes them. Blood always moves from higher

pressure to lower pressure, and a valve is pushed open or shut by the difference in pressure on its two sides.

- The AV valves (tricuspid on the right, mitral or bicuspid on the left) sit between each atrium and its ventricle. When the ventricles are relaxed, pressure is higher in the atria, so the AV valves are open and blood flows into the ventricles.
- When the ventricles contract, ventricular pressure pushes the cusps up until their edges meet and the valve closes. The papillary muscles contract at the same time and pull on the chordae tendineae, so the cusps cannot flip back into the atria.
- The semilunar valves (pulmonary and aortic) sit at the exits of the ventricles. Each has three cup-shaped cusps. They open only when ventricular pressure rises above the pressure in the artery.
- When the ventricles relax, blood in the arteries starts to fall back toward the heart. It fills the cusps, the cusps meet, and the semilunar valve closes.

Fibrous skeleton

Four rings of dense connective tissue around the valves. It holds the valves in place, gives the muscle something to attach to, and acts as an electrical insulator between the atria and the ventricles.

Interventricular septum

The muscular wall between the two ventricles. The AV bundle and the bundle branches run in it.

The path of blood through the heart

Watch: ▶ Flow through the heart (Khan Academy)

1. Blood low in oxygen returns from the body in the superior and inferior venae cavae and enters the right atrium.
2. It passes through the tricuspid valve into the right ventricle.
3. The right ventricle pumps it through the pulmonary valve into the pulmonary trunk and the pulmonary arteries.
4. In the lungs, the blood gives up CO2 and picks up O2.
5. The four pulmonary veins carry it back to the left atrium.
6. It passes through the mitral valve into the left ventricle.
7. The left ventricle pumps it through the aortic valve into the aorta and out to the body.

Coronary circulation

- The heart muscle gets its own blood supply from the right and left coronary arteries, which branch off the aorta just above the aortic valve.
- The coronary veins drain into the coronary sinus, which empties into the right atrium.
- Contracting muscle squeezes the coronary vessels, so the left ventricle gets most of its blood flow while the heart is relaxed, in diastole. A very fast heart rate shortens diastole and leaves less time for coronary flow.

A. Smooth muscle: how it contracts and what controls it

Competencies 1 and 2: Smooth muscle contraction mechanism; Smooth muscle types and regulation.

In your textbook: Silverthorn Chapter 13, the part on smooth muscle.

We start with smooth muscle because the blood vessels in next week's notes are made of it. Skeletal muscle needs a motor neuron and troponin to contract. Smooth muscle needs neither. Its on switch is on the myosin, and an enzyme flips it. That one difference explains why smooth muscle is slow, why it can hold tension for hours, and why it uses so little ATP.

Where the calcium comes from

- Most of the Ca2+ comes in from the extracellular fluid through voltage-gated L-type Ca2+ channels, which open when the membrane depolarizes. Stretching the cell or a signal from a neighboring cell can depolarize it.
- A hormone or neurotransmitter can also raise Ca2+ without any change in membrane voltage. It binds a G protein-coupled receptor, which releases a second messenger called IP3, and IP3 opens Ca2+ channels on the sarcoplasmic reticulum (SR).
- The SR in smooth muscle is small, so smooth muscle depends heavily on Ca2+ from outside the cell. This is why calcium channel blockers relax blood vessels.

Contraction and relaxation, step by step

Watch: ▶ Smooth muscle structure and contraction (RegisteredNurseRN)

1. Ca2+ rises in the cytoplasm.
2. Ca2+ binds calmodulin, a calcium-binding protein. Calmodulin does the job troponin does in skeletal muscle, but it works on myosin instead of actin.
3. The Ca2+-calmodulin complex switches on myosin light chain kinase (MLCK).
4. MLCK adds a phosphate to the myosin light chain, which turns on the myosin head.

5. The myosin heads bind actin and the crossbridges cycle. The actin is anchored to dense bodies, so the whole cell draws in.
6. Ca2+ is pumped out of the cell and back into the SR, and MLCK turns off.
7. Myosin light chain phosphatase removes the phosphate from myosin, the crossbridges stop cycling, and the muscle relaxes.

Dense bodies

Protein anchors where actin attaches. They do the job that Z discs do in skeletal muscle.

Latch state

Myosin heads that lose their phosphate while still bound to actin can stay attached and keep holding tension with almost no ATP. This is how sphincters and artery walls hold their tone for hours.

Why smooth muscle is slower and uses less energy

- The switch is on myosin and needs an enzyme step, which takes time.
- Smooth muscle myosin breaks down ATP slowly, so its crossbridges cycle much more slowly than in skeletal muscle.
- The latch state holds tension for very little ATP.
- There are no sarcomeres, so a smooth muscle cell can shorten much farther than a skeletal muscle fiber.

Single-unit and multi-unit smooth muscle

	Single-unit	Multi-unit
How the cells connect	Gap junctions link the cells, so the whole sheet contracts together	Few gap junctions; each cell has its own nerve supply
Pacemaker cells	Some cells set a rhythm	None
Control	The tissue works as one unit	Fine, graded control
Examples	Gut, uterus, bladder, small blood vessels	Iris of the eye, large airways, large arteries

Many tissues have features of both types, so these two labels are a simplification.

What changes smooth muscle tone

Signal	Example	What it does
Stretch	Rising pressure stretches an arteriole	The muscle contracts, which keeps flow steady
Local chemicals	Low O2, high CO2, H+, nitric oxide	Relax arterioles, so an active tissue gets more blood
Autonomic nerves	Norepinephrine on alpha-1 receptors	Contracts the smooth muscle of blood vessels
Hormones	Epinephrine on beta-2 receptors	Relaxes the smooth muscle of airways and of some blood vessels
Hormones	Angiotensin II, vasopressin	Contract blood vessels

The receptor decides the response, not the chemical. Epinephrine contracts vessels that have alpha-1 receptors and relaxes vessels and airways that have beta-2 receptors.

B. Cardiac muscle cells, and how they compare

Competencies 3 and 4: Cardiac muscle properties; Comparison of the three muscle types.

In your textbook: Silverthorn Chapter 13, the part on cardiac muscle, and Chapter 15, the part on cardiac muscle.

What a cardiac muscle cell looks like

Watch: ▶ Contractile cells of the heart · ▶ Cardiac muscle fibers, the micro anatomy

- Cardiac muscle cells are shorter than skeletal muscle fibers and branched, with a stair-step look.
- Each cell usually has one nucleus in the center, sometimes two.
- Neighboring cells are joined end to end by intercalated discs.
- Cardiac cells have the same actin, myosin, Z discs and bands as skeletal muscle, so they are striated.
- Mitochondria are larger and far more numerous than in skeletal muscle. They fill about 25% of the cell, compared with about 2% in skeletal muscle.
- The T-tubules are wider and fewer, about one per sarcomere at the Z disc. The SR is smaller, so the cell holds a smaller store of Ca2+ and depends on Ca2+ that enters from outside.

Intercalated disc

The thickened junction where two cardiac cells meet. It holds two structures.

Desmosomes

Hold the cells together, so they do not pull apart when the heart contracts.

Gap junctions

Channels that let ions pass from one cell to the next, so an action potential spreads from cell to cell and the heart contracts as a unit.

How the heart makes its ATP

- Cardiac muscle makes its ATP almost entirely by aerobic respiration in its many mitochondria, so it needs a steady supply of oxygen from the coronary arteries. Myoglobin inside the cells holds a small reserve of O2.
- At rest, the heart burns mostly fatty acids (about 60%) and glucose (about 35%), plus smaller amounts of lactic acid, amino acids and ketone bodies.
- During exercise it uses more lactic acid, made by the working skeletal muscles.

The three muscle types compared

Watch: ▶ Skeletal and cardiac muscle: how they are alike · ▶ Skeletal and cardiac muscle: how they differ, part 1 · ▶ Skeletal and cardiac muscle: how they differ, part 2

	Skeletal	Cardiac	Smooth
Look	Long, striated, many nuclei	Branched, striated, one nucleus, intercalated discs	Spindle shaped, no striations, one nucleus
Control	Voluntary, motor neurons	Involuntary; sets its own rhythm, adjusted by autonomic nerves and hormones	Involuntary; autonomic nerves, hormones, stretch and local chemicals
Signal that starts contraction	Acetylcholine from a motor neuron	Action potential from the pacemaker, through gap junctions	Many: stretch, nerves, hormones, pacemaker cells
Where the Ca2+ comes from	SR	Some from outside the cell, which triggers more from the SR	Mostly from outside the cell
Ca2+ switch	Troponin on actin	Troponin on actin	Calmodulin and MLCK on myosin
Gap junctions	None	Yes	Yes in single-unit
Can it summate to tetanus?	Yes	No, the long refractory period prevents it	Holds tone in the latch state

C. The adrenal medulla: a sympathetic ganglion that releases hormones

Competency 5: Adrenal medulla.

In your textbook: Silverthorn Chapter 12, the part on the adrenal medulla.

Most sympathetic pathways use two neurons in a row. In the adrenal medulla, the second neuron is replaced by a gland cell that releases its chemical into the blood instead of onto one target. Because the signal travels in the blood, it reaches the whole body, including the heart, and it lasts longer.

The pathway, step by step

Watch: ▶ Adrenal medulla, adrenaline and noradrenaline (Khan Academy)

1. A sympathetic neuron in the thoracic spinal cord fires. Its axon runs to the adrenal medulla without a synapse along the way.
2. It releases acetylcholine onto nicotinic receptors on the chromaffin cells.

3. The chromaffin cells are modified postganglionic neurons with no axons. They release about 80% epinephrine and 20% norepinephrine into the blood.
4. The blood carries these hormones to every tissue with adrenergic receptors. In the heart they bind beta-1 receptors and raise both heart rate and the force of contraction.
5. The effect lasts minutes, because the hormones are cleared from the blood slowly.

	Sympathetic nerve	Adrenal medulla
Chemical	Norepinephrine, released onto the target	Mostly epinephrine, carried in the blood
Reach	Only the tissue the nerve supplies	Every tissue with adrenergic receptors
Speed and length	Fast and brief	Slower to arrive and lasts minutes

D. The pacemaker and the cardiac action potential

Competencies 6, 7 and 9: Pacemaker potential; Contractile cell action potential; Cardiac refractory period.

In your textbook: Silverthorn Chapter 15, the part on electrical signals in cardiac muscle.

The heart has two kinds of muscle cells. About 1% are autorhythmic cells. They make their own action potentials without any signal from a nerve, which is why a transplanted heart keeps beating even though all of its nerves were cut. The rest are contractile cells, which do the pumping once an action potential reaches them.

Autorhythmic cells do two jobs

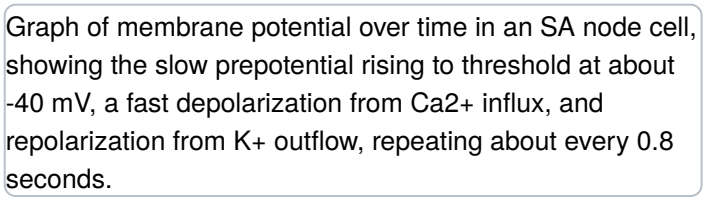
1. They act as the pacemaker and set the rhythm of the heartbeat.
2. They form the conduction system, the path each action potential follows so the chambers contract in the right order. Abnormal conduction or contraction is called an arrhythmia.

The pacemaker potential in the SA node, step by step

Watch: ▶ What sets the pacemaker rate of the heart? · ▶ The pacemaker potential · ▶ The depolarization phase of the action potential · ▶ The repolarization phase of the action potential

An SA node cell has no stable resting potential. Its membrane drifts up on its own after every beat, which is what makes it a pacemaker.

1. After an action potential, the membrane is at its most negative, about -60 mV.
2. Channels called funny channels (If) open and let Na+ leak in. The membrane slowly depolarizes. This slow climb is the pacemaker potential.
3. As the membrane nears threshold, some Ca2+ channels open and speed the climb.
4. At threshold, about -40 mV, voltage-gated Ca2+ channels open and Ca2+ rushes in. In pacemaker cells, the rising phase of the action potential comes from Ca2+, not Na+.
5. K+ channels open, K+ leaves the cell, and the membrane repolarizes back to about -60 mV. The funny channels open again and the cycle repeats.



The SA node action potential. The slow climb (prepotential) comes from Na+ leaking in, the rise from Ca2+ rushing in, and the fall from K+ leaving. OpenStax College, CC BY 3.0. Source

The faster the pacemaker potential climbs, the sooner the cell reaches threshold and the faster the heart beats. This is how the autonomic nerves change heart rate, which is covered in part G.

The action potential in a contractile cell, step by step

Watch: ▶ How a typical cardiac cell is activated, in order · ▶ Action potentials in cardiac myocytes (Khan Academy)

A contractile cell has a stable resting membrane potential of about -90 mV. It reaches threshold only when a neighboring cell passes it current through gap junctions.

1. Depolarization: voltage-gated fast Na+ channels open, Na+ rushes in, and the membrane rises quickly. The fast Na+ channels then close.
2. Plateau: voltage-gated slow Ca2+ channels open and Ca2+ flows in from the extracellular fluid. At the same time some K+ channels let K+ out. Ca2+ in balances K+ out, so the membrane stays near 0 mV for about 0.25 seconds. Skeletal muscle has no plateau.

3. The Ca2+ that enters triggers the SR to release much more Ca2+. This is called calcium-induced calcium release.
4. Ca2+ binds troponin, the actin and myosin filaments slide past each other, and the cell develops tension.
5. Repolarization: the Ca2+ channels close and more K+ channels open, so K+ leaves and the membrane returns to -90 mV. Ca2+ is pumped back into the SR and out of the cell, and the muscle relaxes.

Anything that changes Ca2+ entry through the slow Ca2+ channels changes the strength of contraction. Epinephrine increases Ca2+ entry, so the heart contracts more forcefully.

The refractory period

- The refractory period is the time when a second action potential cannot be started.
- In cardiac muscle it lasts almost as long as the contraction itself, because of the plateau. The muscle has started to relax before another action potential is possible.
- So cardiac muscle cannot summate or go into tetanus. A heart held in a sustained contraction could not fill, and blood flow would stop.

Graph of a contractile cardiac cell action potential showing rapid depolarization, the long plateau, and repolarization, with the absolute and relative refractory periods marked across most of the action potential, and a comparison of skeletal and cardiac muscle.

The contractile cell action potential, with its long plateau and long refractory period.
OpenStax College, CC BY 3.0. Source

E. The conduction system and the EKG

Competencies 8, 10 and 11: Conduction system; ECG waves and intervals; ECG interpretation.

In your textbook: Silverthorn Chapter 15, the parts on electrical conduction in the heart and the electrocardiogram.

The path of each action potential

Watch: ▶ EKG lecture: the heart's electrical pathway (9:13) · ▶ Electrical system of the heart (Khan Academy)

1. SA node: in the wall of the right atrium, just below where the superior vena cava enters. It fires first and the signal spreads across both atria, which contract.
2. AV node: in the floor of the right atrium, in the septum between the atria, near the opening of the coronary sinus. It slows the signal down by about 0.1 second, which gives the atria time to empty into the ventricles.
3. AV bundle (bundle of His): the only electrical path from the atria to the ventricles. Everywhere else, the fibrous skeleton blocks the signal.
4. Right and left bundle branches: run down the interventricular septum toward the apex.
5. Purkinje fibers: spread from the apex up into the walls of the ventricles. The ventricles contract from the apex upward, squeezing blood toward the semilunar valves.

Front view of the heart with the conduction system drawn in yellow: the SA node in the right atrium, the internodal pathways, the AV node, the AV bundle, the right and left bundle branches, and the Purkinje fibers in the ventricle walls.

The conduction system of the heart.
OpenStax College, CC BY 3.0. Source

Why the SA node is in charge

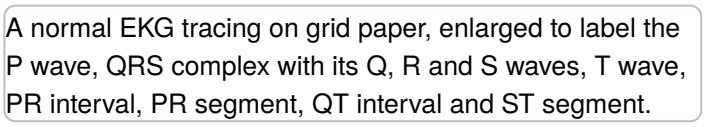
Part of the system	Rate it would set on its own
SA node	About 100 beats per minute
AV node	About 40 to 60 beats per minute
AV bundle, bundle branches and Purkinje fibers	About 20 to 40 beats per minute

The SA node fires fastest, so its signal reaches every other autorhythmic cell and triggers it before that cell can fire at its own slower rate. That makes the SA node the natural pacemaker. If it fails, a slower part of the system takes over. Nerves (the autonomic nervous system) and hormones such as epinephrine change the timing and strength of each beat, but they do not create the rhythm.

What an EKG records

Watch: ▶ EKG lecture: overview of the EKG (start here) · ▶ What is an EKG?

- An electrocardiogram (EKG or ECG) is a recording of the electrical activity of the whole heart during each beat, picked up by electrodes on the skin.
- A 12-lead EKG combines limb and chest electrodes to make 12 tracings, each looking at the heart from a different angle.
- An EKG can show an abnormal conduction pathway, an enlarged heart, damaged areas of the heart, and the cause of some chest pain. It is always compared with the patient's earlier tracings.



A normal EKG, with each wave, interval and segment labeled.
OpenStax College, CC BY 3.0. Source

The waves

Watch: ▶ EKG lecture: electrical and mechanical events (17:41) · ▶ EKG lecture: the EKG waves and the cardiac cycle (24:28) · ▶ The EKG waves and what each one means · ▶ The isoelectric line

P wave

A small upward bump: the atria depolarizing, as the signal spreads from the SA node across the atria.

QRS complex

A downward dip, a tall upward spike, and another dip: the ventricles depolarizing as the signal spreads through them.

T wave

A rounded upward bump: the ventricles repolarizing, just as they start to relax.

Where is atrial repolarization?

It happens during the QRS complex, which is much larger and hides it.

Intervals and segments

Watch: ▶ EKG lecture: key EKG intervals to know (37:55) · ▶ Why the ST segment matters · ▶ Why the QT interval matters

PR (PQ) interval

From the start of the P wave to the start of the QRS. It is the time for the signal to cross the atria, the AV node and the rest of the conduction system. Normal is 0.12 to 0.20 seconds. It gets longer when the signal has to detour around scar tissue, for example after coronary artery disease or rheumatic fever.

QRS duration

How long the ventricles take to depolarize. Normal is under 0.12 seconds.

ST segment

From the end of the S wave to the start of the T wave, while the ventricular cells are in their plateau. It should be flat. ST elevation suggests an acute heart attack (myocardial infarction). ST depression suggests ischemia, meaning the heart muscle is not getting enough blood flow.

QT interval

From the start of the QRS to the end of the T wave: ventricular depolarization and repolarization together. It can be abnormal with heart muscle damage, ischemia or conduction problems.

When the waves change size

Watch: ▶ Abnormalities an EKG can show

Change	What it can mean
Large P wave	Enlarged atrium
Enlarged Q wave	A past heart attack
Large R wave	Enlarged ventricles
Flat T wave	The heart muscle is not getting enough oxygen, as in coronary artery disease
Tall, peaked T wave	High blood potassium (hyperkalemia)

Reading the heart rate from a strip

Watch: ▶ EKG lecture: measuring on the EKG grid (32:13)

- EKG paper runs at 25 mm per second. One small box is 0.04 seconds, one big box (five small boxes) is 0.20 seconds, and five big boxes are one second.
- Regular rhythm: divide 300 by the number of big boxes between two R waves.
- Irregular rhythm: count the QRS complexes in six seconds and multiply by 10.

Two tests that use the EKG

Watch: ▶ What is a stress test? · ▶ What is a Holter monitor?

Stress test

An EKG recorded during exercise. Narrowed coronary arteries may carry enough blood at rest but not during exercise, so the EKG changes only when the heart is working hard.

Holter monitor

A portable EKG worn for about 24 hours. It catches abnormal rhythms or episodes of poor blood flow that come and go.

Matching the EKG to the contraction, step by step

Watch: ▶ Conduction system, ventricular contraction and the ECG (Armando Hasudungan)

Each electrical event comes just before the contraction it causes. At a heart rate of 75 beats per minute:

1. The action potential starts in the SA node and spreads across the atria. The atrial cells depolarize, and the P wave appears.
2. Just after the P wave begins, the atria contract (atrial systole). The signal slows down at the AV node for about 0.1 second, so the atria have time to finish filling the ventricles.
3. The signal speeds through the AV bundle, the bundle branches and the Purkinje fibers, and the ventricles depolarize. This makes the QRS complex. The atria repolarize at the same time, hidden inside it.
4. The ventricles contract (ventricular systole) shortly after the QRS appears and keep contracting through the ST segment. Contraction moves from the apex toward the base, squeezing blood up toward the semilunar valves.
5. The ventricles repolarize, starting at the apex. This makes the T wave.
6. As the T wave ends, the ventricles relax (ventricular diastole).

Six drawings of the heart in a circle, each showing how far the electrical signal has spread, next to an EKG with the matching part of the tracing highlighted, from the SA node firing through ventricular repolarization.

How each part of the EKG matches the spread of the signal through the heart.
OpenStax College, CC BY 3.0. Source

F. The cardiac cycle, the pressure-volume loop, and heart sounds

Competencies 13, 14, 15 and 21: Cardiac cycle; Pressure volume loop; Heart sounds; Heart sound and pulse correlation.

In your textbook: Silverthorn Chapter 15, the part on the heart as a pump.

Watch: ▶ One cardiac cycle, with the EKG waves and the mechanical events · ▶ Systole and diastole

One cardiac cycle is everything that happens in one heartbeat. The atria and ventricles each contract and relax, and blood always moves from higher pressure to lower pressure. Systole is the contraction phase of a chamber, and diastole is its relaxation phase. The left and right sides run at very different pressures, but they eject the same volume of blood with each beat.

A circular diagram of one cardiac cycle starting with atrial contraction, then isovolumetric contraction, ventricular ejection, isovolumetric relaxation and ventricular filling, with a small EKG beside each phase.

The phases of one cardiac cycle, with the part of the EKG that goes with each one.
OpenStax College, CC BY 3.0. Source

One cycle at 75 beats per minute, step by step

Watch: ▶ Cardiac cycle, step 1: ventricular filling · ▶ End-diastolic volume and what sets it · ▶ Cardiac cycle, step 2: isovolumetric contraction · ▶ Cardiac cycle, step 3: ventricular ejection · ▶ End-systolic volume · ▶ Cardiac cycle, step 4: isovolumetric relaxation · ▶ The cardiac cycle, all together · ▶ Important points about the cardiac cycle · ▶ The left heart and the right heart

At this rate one cycle takes 0.8 seconds. The numbers below are for the left ventricle. The right side follows the same steps at lower pressures.

Atrial systole (about 0.1 second): the atria contract while the ventricles stay relaxed.

1. The SA node fires and the atria depolarize. This is the P wave.
2. The atria contract and push blood through the open AV valves into the ventricles.
3. This adds about the last 25 mL to the roughly 105 mL already in each ventricle. Each ventricle now holds about 130 mL. This is the end-diastolic volume (EDV).
4. The QRS complex marks the start of ventricular depolarization.

Ventricular systole (about 0.3 second): the ventricles contract while the atria relax.

1. The ventricles start to contract and their pressure rises above atrial pressure. Blood pushes the AV valves shut.
2. For about 0.05 second, all four valves are closed. The muscle is squeezing, but no blood can leave, so the volume stays the same. This is isovolumetric contraction.
3. When pressure in the left ventricle rises above the pressure in the aorta (about 80 mmHg), and the right ventricle rises above the pressure in the pulmonary trunk (about 20 mmHg), the semilunar valves open. This is ventricular ejection, and it lasts about 0.25 second.
4. Pressure in the left ventricle peaks at about 120 mmHg, and in the right ventricle at about 25 to 30 mmHg.
5. Each ventricle ejects about 70 mL. About 60 mL stays behind. This is the end-systolic volume (ESV).

6. The T wave marks the start of ventricular repolarization.

Relaxation period (about 0.4 second): the atria and ventricles are both relaxed.

1. The ventricles relax and their pressure falls. Blood in the aorta and pulmonary trunk starts to flow back toward the ventricles, fills the semilunar cusps, and closes the valves. The aortic valve closes at about 100 mmHg.
2. Blood bouncing off the closed aortic valve makes a small bump on the aortic pressure curve called the dicrotic notch.
3. With all four valves closed again, ventricular volume does not change. This is isovolumetric relaxation.
4. Ventricular pressure keeps falling until it drops below atrial pressure. The AV valves open and the ventricles fill passively. By the end of the relaxation period they are about three quarters full.
5. The next P wave appears, and a new cycle begins.

Stroke volume (SV)

The volume each ventricle ejects in one beat. SV = EDV - ESV. In this example, 130 mL - 60 mL = 70 mL.

Ejection fraction (EF)

The share of the EDV that is ejected. EF = SV ÷ EDV × 100. Normal is about 55 to 70 percent.

Books give slightly different typical volumes. The Week 10 lab uses an EDV of 120 mL and an ESV of 50 mL, which also gives a stroke volume of 70 mL.

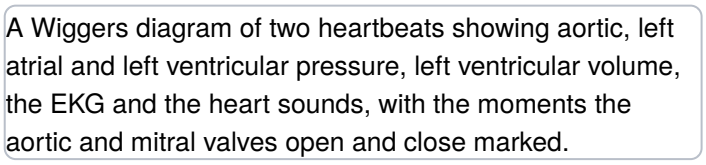
When the heart beats faster

A faster heart rate shortens the relaxation period much more than it shortens systole. Less relaxation time means less time for the ventricles to fill and less time for blood to flow through the coronary arteries.

The Wiggers diagram

Watch: ▶ Pressure in the left heart, part 1 (Khan Academy)

A Wiggers diagram stacks the pressures, the ventricular volume, the EKG and the heart sounds on one time line, so you can see what happens at the same moment in each. Read it from top to bottom at any one time point.



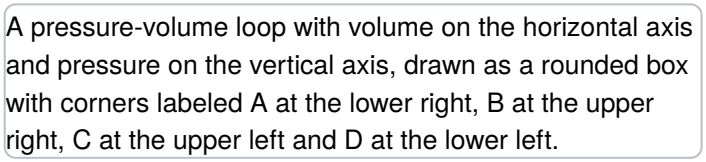
A Wiggers diagram for the left side of the heart. The valves open and close where the pressure lines cross.
adh30, revised by DanielChangMD from work by DestinyQx, redrawn as SVG by xavax. CC BY-SA 4.0. Source

- The mitral valve closes where ventricular pressure rises above atrial pressure. The first heart sound comes here, just after the QRS.
- The aortic valve opens where ventricular pressure rises above aortic pressure. Ventricular volume starts to fall.
- The aortic valve closes where ventricular pressure falls below aortic pressure. The second heart sound and the dicrotic notch come here, near the end of the T wave.
- The mitral valve opens where ventricular pressure falls below atrial pressure. Ventricular volume starts to rise again.

The pressure-volume loop

Watch: ▶ Understanding the pressure-volume loop (Khan Academy) · ▶ Drawing a pressure-volume loop (Khan Academy)

A pressure-volume loop plots left ventricular pressure against left ventricular volume for one beat. Each side of the loop is one phase of the cycle.



A pressure-volume loop for the left ventricle. Read it counterclockwise from A.
Dhollm, CC BY-SA 3.0. Source

1. A to B, isovolumetric contraction: the mitral valve has just closed at A. Pressure rises sharply but volume stays at the EDV.
2. B to C, ejection: the aortic valve opens at B. Blood leaves, so volume falls while pressure rises and then falls.
3. C to D, isovolumetric relaxation: the aortic valve closes at C. Pressure falls sharply but volume stays at the ESV.
4. D to A, filling: the mitral valve opens at D. Blood flows in, so volume rises while pressure stays low.

- The width of the loop is the stroke volume (EDV minus ESV).
- A stronger contraction (more contractility) lets the ventricle empty further, so the loop gets wider toward the left.

- Higher afterload, such as high blood pressure, means the ventricle has to build more pressure before the aortic valve opens, so the loop gets taller and usually narrower.

Heart sounds

Watch: ▶ What is auscultation? · ▶ The typical heart sounds · ▶ Valve closure, heart sounds and the pressure changes · ▶ Where to listen for each heart sound · ▶ The normal heart sounds and the phases they belong to · ▶ Heart murmurs · ▶ Fixed split second heart sound, with a real patient (Stanford Medicine 25) · ▶ The exam for diastolic murmurs, with a real patient (Stanford Medicine 25) · ▶ The exam for aortic regurgitation, with a real patient (Stanford Medicine 25)

Auscultation means listening to sounds inside the body, usually with a stethoscope. Smoothly flowing blood is silent. The heart sounds come from the turbulence of blood when the valves snap shut. Each cycle has four heart sounds, but in a normal adult heart only the first two are loud enough to hear.

S1, lub

Louder and a little longer. The AV valves closing, just after ventricular systole starts and just after the QRS.

S2, dub

Shorter and softer. The semilunar valves closing, at the start of ventricular diastole, near the end of the T wave.

S3

A soft sound early in diastole, from blood rushing into the ventricles. It is normal in children and athletes. In an older adult it can mean a failing ventricle that is overfilled.

S4

A sound late in diastole, when the atria push blood into a stiff ventricle, as in long-standing high blood pressure.

- At a normal heart rate, the gap from S1 to S2 (systole) is shorter than the gap from S2 to the next S1 (diastole). That is how you tell the two sounds apart.
- Each valve is heard best at a spot on the chest a little away from the valve itself, because the sound is carried by the blood flowing away from it.
- The pulse you feel at the wrist comes just after S1. S1 marks the start of ventricular contraction, and ejection follows a moment later.
- A murmur is the sound of turbulent flow through a valve that does not work right. Stenosis is a valve that is too narrow and does not open fully. Insufficiency (regurgitation) is a valve that does not close completely, so blood leaks backward. Mitral valve prolapse, where a cusp bulges back into the left atrium, is a common cause of mitral insufficiency.

Graph of aortic, left ventricular and left atrial pressure over one cycle, marking where the AV valves close and open and where the semilunar valves open and close, above a heart sound tracing labeled first (lub), second (dub) and third.

The heart sounds lined up with the pressures. S1 comes when the AV valves close and S2 when the semilunar valves close.
OpenStax College, CC BY 3.0. Source

G. Stroke volume, cardiac output, and what controls them

Competencies 16, 17, 18, 19 and 20: Stroke volume and ejection fraction; Cardiac output; Frank Starling relationship; Preload afterload and contractility; Autonomic regulation of the heart.

In your textbook: Silverthorn Chapter 15, the part on cardiac output.

Watch: ▶ Cardiac output and what determines it · ▶ Cardiac output (Armando Hasudungan)

The body needs a certain amount of oxygen from the blood every minute. Cells that are working hard, as in exercise, take up more oxygen. Cells at rest need less. The heart matches its output to that need.

Cardiac output (CO)

The volume of blood each ventricle pumps in one minute.

Stroke volume (SV)

The volume each ventricle pumps in one beat.

Heart rate (HR)

The number of beats per minute.

CO = SV × HR. For a typical resting adult, 70 mL per beat × 75 beats per minute = 5,250 mL per minute, or about 5.25 L per minute. That is close to the whole blood volume (about 5 L), so all of your blood passes through the heart about once a minute.

Cardiac output rises when stroke volume or heart rate rises

Watch: ▶ Cardiac reserve

Situation	Stroke volume	Heart rate	Cardiac output
Rest	70 mL	75 beats/min	5.25 L/min
Mild exercise	100 mL	100 beats/min	10 L/min
Intense exercise	130 mL	150 beats/min	19.5 L/min

Exercise, fever and thyroid hormone all raise cardiac output.

Cardiac reserve

The difference between a person's maximum cardiac output and their resting cardiac output. In a healthy adult the maximum is about 4 to 5 times the resting output. In top endurance athletes it can be 7 to 8 times. In heart failure there is little or no reserve, so even daily activities can be hard.

Three things set the stroke volume

Watch: ▶ Three factors that set stroke volume and keep both ventricles equal

In a healthy heart, whatever blood comes in during diastole is pumped out during the next systole. If more blood comes in, more goes out. Three factors set stroke volume and keep the two ventricles pumping the same amount:

1. Preload: how much the ventricle is stretched before it contracts.
2. Contractility: how forcefully the ventricular muscle contracts.
3. Afterload: the pressure the ventricle has to overcome before blood can leave.

A bicycle is a useful comparison for all of these. The speed of the bike is the cardiac output.

Preload and the Frank-Starling law

Watch: ▶ What is preload? (Khan Academy) · ▶ Frank-Starling mechanism (Khan Academy)

- Preload is how stretched the ventricle is just before it contracts. It goes up and down with the end-diastolic volume.
- The Frank-Starling law of the heart: the more the muscle is stretched (within limits), the more forcefully it contracts. The more the ventricle fills in diastole, the more it pumps out in systole.
- A rubber band works the same way. The farther you stretch it, the harder it snaps back.
- On the bicycle, preload is a tailwind. It lets the rider go faster with no extra effort. Unlike the wind, the heart has a maximum: past a certain point, more stretch no longer adds force.

Two things decide the end-diastolic volume:

1. How long the ventricles have to fill. A faster heart rate shortens filling time, so EDV falls.
2. Venous return, the volume of blood coming back to the heart. More venous return means more filling, so EDV rises.

Contractility

Watch: ▶ Substances that change contraction: inotropy and chronotropy

- Contractility is the strength of contraction at any given preload.
- Positive inotropes make contraction stronger, mainly by letting more Ca2+ in during the action potential. Examples are the sympathetic nerves, epinephrine and norepinephrine, more Ca2+ in the extracellular fluid, and the drug digitalis.
- Negative inotropes make contraction weaker. Examples are less sympathetic activity, lack of oxygen, acidosis, some anesthetics, high K+ in the extracellular fluid, and calcium channel blockers.
- On the bicycle, contractility is how hard the rider pushes on the pedals. Pushing harder makes the bike go faster. Too little push slows it down.

Afterload

Watch: ▶ What is afterload? (Khan Academy) · ▶ Summary tables: what changes cardiac output

- Afterload is the pressure the ventricle must overcome to open the semilunar valve and eject blood. It depends mostly on arterial blood pressure.
- High blood pressure and arteries narrowed by atherosclerosis both raise afterload, so less blood is ejected with each beat.
- On the bicycle, afterload is the road. A smooth road or a gentle downhill lets the bike go faster for the same effort. A narrow, bumpy road slows it down. Lowering afterload can raise cardiac output, especially when contractility is weak, as in heart failure.

Flow chart showing that autonomic innervation, hormones, the atrial reflex and venous return set heart rate, and that venous return, filling time, autonomic innervation, hormones and vasodilation or vasoconstriction act through preload, contractility and afterload to set EDV and ESV and so stroke volume, with cardiac output equal to heart rate times stroke volume.

Everything that changes cardiac output works through heart rate or stroke volume.
OpenStax College, CC BY 3.0. Source

Heart rate

- Positive chronotropes speed up the heart rate. Negative chronotropes slow it down.
- On the bicycle, heart rate is how fast the rider pedals. Faster pedaling makes the bike go faster, up to a point.
- If the heart beats too slowly, cardiac output drops. If it beats too fast, cardiac output also drops, because the ventricles no longer have enough time to fill between beats.

How the nervous system sets heart rate

Watch: ▶ Autonomic nervous system effects on the heart (Khan Academy)

Left alone, the SA node would fire about 100 times a minute. The cardiovascular center in the medulla oblongata adjusts that rate through the sympathetic and parasympathetic nerves, using input from three kinds of sensors:

Proprioceptors

Sense the position of the limbs and muscles. They are the main reason heart rate rises quickly when you start to move.

Chemoreceptors

Sense chemical changes in the blood.

Baroreceptors

Sense the stretch of the large arteries caused by blood pressure. The most important ones are in the aortic arch and the carotid arteries.

Sympathetic pathway, step by step:

1. The cardiovascular center in the medulla sends signals down sympathetic neurons in the thoracic spinal cord.
2. The cardiac accelerator nerves carry them to the SA node, the AV node and most of the heart muscle.
3. The nerve endings release norepinephrine, which binds beta-1 receptors.
4. In the SA and AV nodes, norepinephrine makes the pacemaker potential climb faster, so the cells reach threshold sooner and heart rate goes up.
5. In the contractile cells, norepinephrine lets more Ca2+ in through the slow Ca2+ channels, so each contraction is stronger.

Parasympathetic pathway, step by step:

1. Signals travel from the medulla in the right and left vagus nerves (cranial nerve X).
2. The vagal fibers end on the SA node, the AV node and the atrial muscle.
3. They release acetylcholine, which slows the pacemaker potential, so heart rate goes down.
4. Very few vagal fibers reach the ventricles, so the parasympathetic system has little effect on contractility.

The heart is always under both systems, and the balance shifts constantly. At rest the parasympathetic side wins, which is why the resting heart rate is about 75 beats per minute instead of the SA node's own 100. Strong vagal stimulation can slow the heart to 20 or 30 beats per minute.

Chemicals that change heart rate

- Low oxygen (hypoxia), acidosis and alkalosis all depress heart activity.
- Epinephrine and norepinephrine from the adrenal medulla raise heart rate and contractility. Exercise, stress and excitement trigger their release.
- Thyroid hormone raises heart rate and contractility. A resting heart rate over 100 (tachycardia) is one sign of hyperthyroidism.
- High blood K+ blocks the generation of action potentials, so heart rate and force fall. High blood Na+ blocks Ca2+ entry during the action potential, so force falls.
- A moderate rise in Ca2+ in the extracellular fluid speeds the heart rate and strengthens the beat.

Other things that affect heart rate

Factor	Effect on resting heart rate
Age	Slows with age. Newborns are around 120 beats per minute.
Sex	Adult women have a slightly higher resting rate.
Fitness	Slows with training. A trained athlete can have a resting rate below 50, which is a healthy, efficient bradycardia.
Body temperature	Fever and strenuous exercise raise it. A low body temperature slows the rate and weakens the beat. Surgeons sometimes cool a patient on purpose during heart surgery, because a cold body needs less oxygen and tolerates reduced blood flow longer.

When the pump fails

In your textbook: Silverthorn Chapter 15, the part on cardiac output, and Chapter 16, the part on cardiovascular disease.

Heart failure

Watch: ▶ Left-sided heart failure · ▶ Right-sided heart failure · ▶ BNP and heart failure

- In heart failure, the heart cannot pump enough blood to meet the body's needs. Ejection fraction is often low, and cardiac reserve is small.
- Left-sided failure: the left ventricle cannot keep up, so blood backs up into the lungs. The patient is short of breath and may not be able to lie flat.
- Right-sided failure: the right ventricle cannot keep up, so blood backs up into the body's veins. The patient has swollen ankles and legs.
- A stretched, overloaded ventricle releases BNP (brain natriuretic peptide). A high BNP in the blood supports a diagnosis of heart failure.

What leaks out when heart cells are injured

Watch: ▶ Cardiac enzymes and an acute MI

When heart muscle cells are damaged, as in a heart attack, proteins that normally stay inside the cells leak into the blood. Each one rises and falls on its own schedule.

Marker	Comes from	Rises	Back to normal	What it is good for
Myoglobin	Any striated muscle	1 to 3 hours	About 24 hours	The earliest sign, but not specific to the heart
CK-MB	Mostly heart muscle	3 to 6 hours	2 to 3 days	Spotting a second heart attack in the days after the first
Troponin I or T	Heart muscle only	3 to 6 hours	7 to 14 days	The standard test for heart muscle injury

Creatine kinase (CK)
: The enzyme that moves a phosphate from creatine phosphate to ADP to make ATP. It normally stays inside cells. CK-MM comes from skeletal muscle, CK-BB from the brain, and CK-MB from heart muscle.

In a person at rest, heart muscle can stay alive on as little as 10 to 15% of its normal blood supply.

Aging and congenital heart defects

Watch: ▶ How the heart changes with age

Congenital heart defects are present at birth. Most are a hole or a narrowing that changes where blood flows.

Watch: ▶ Patent foramen ovale · ▶ Patent ductus arteriosus · ▶ Coarctation of the aorta · ▶ Tetralogy of Fallot · ▶ Ventricular septal defect

Patent foramen ovale
: The opening between the two atria that every fetus has fails to seal after birth.

Patent ductus arteriosus
: The fetal vessel that connects the pulmonary trunk to the aorta stays open after birth, so some blood from the aorta flows back into the pulmonary trunk.

Coarctation of the aorta
: A narrowed segment of the aorta. Blood pressure is high in the arms and low in the legs.

Tetralogy of Fallot
: Four defects together: a ventricular septal defect, an aorta that sits over both ventricles, a narrowed pulmonary valve or outflow, and a thickened right ventricle.

Ventricular septal defect
: A hole in the interventricular septum, so blood from the left ventricle leaks into the right ventricle.